Sub Code:23CE/EE/ME/EC/CS/IT/AD/AM1T02 **MIC23**

B.Tech I Semester Regular Examinations

LINEAR ALGEBRA AND CALCULUS

(CIVIL, EEE, MECH, ECE, CSE, IT, AIDS, AIML)

(Model Paper-I)

Time: 3 hours **Max Marks: 70M**

Note: 1. Question paper consists of two parts (**Part A and Part B**)
2. Answer all sub questions from **Part A**
3. Answer all questions from **Part B** with **either or choice**

		PART – A			**10 X 2 = 20 M**
			CO	**BTL**	**Marks**
I.	**a)**	Find the rank of the matrix $\begin{bmatrix}1 & 2 & 3\\3 & 4 & 5\\4 & 5 & 6\end{bmatrix}$	CO1	L1	**2 M**
	b)	Explain the Normal form of a matrix	CO1	L2	**2 M**
	c)	State Cayley-Hamilton Theorem	CO2	L2	**2 M**
	d)	Write the matrix of the quadratic form $4x^2 - 2y^2 + z^2 - 2xy + 6zx$	CO2	L1	**2 M**
	e)	Define Rolle's theorem	CO3	L1	**2 M**
	f)	Write the Maclaurin's series of e^x	CO3	L2	**2 M**
	g)	Define Jacobian of u,v w.r.to x,y	CO4	L1	**2 M**
	h)	Explain Conditions of maximum & minimum of two variable function	CO4	L2	**2 M**
	i)	Evaluate $\int_0^2\int_0^x y\,dy\,dx$	CO5	L5	**2 M**
	j)	Find $\int_0^1\int_1^2\int_2^3 xyz\,dx\,dy\,dz$	CO5	L1	**2 M**
		PART – B			**5 X 10 = 50 M**
		UNIT – I			
1.	**a)**	Find the Inverse of the following matrices by Gauss-Jordan Method $\begin{bmatrix}1 & 1 & 3\\1 & 3 & -3\\-2 & -4 & -4\end{bmatrix}$	CO1	L1	**5 M**
	b)	Determine the rank of the following matrices by reducing to the Echelon form $\begin{bmatrix}0 & 1 & -3 & -1\\1 & 0 & 1 & 1\\3 & 1 & 0 & 2\\1 & 1 & -2 & 0\end{bmatrix}$	CO1	L5	**5 M**
		(OR)			
2.	**a)**	Solve the following system of equations by Gauss-Elimination method x + y + z = 9, 2x – 3y + 4z = 13, 3x + 4y + 5z = 40	CO1	L3	**5 M**
	b)	Test for consistency and solve the following system of equations 5x + 3y + 7z = 4, 3x + 26y + 2z = 9, 7x + 2y + 10z = 5	CO1	L4	**5 M**
		UNIT – II			
3.	**a)**	Verify the Cayley – Hamilton theorem for the following	CO2	L2	**5 M**

		matrices also find A^{-1} for $\begin{bmatrix} 1 & 1 & 3 \\ 1 & 3 & -3 \\ -2 & -4 & -4 \end{bmatrix}$			
	b)	Determine the Eigen values and Eigen vectors of the following matrices $\begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}$	CO2	L5	**5 M**
(OR)					
4.		Reduce the Quadratic form $x^2 + 5y^2 + z^2 + 2xy + 2yz + 6zx$ to canonical form by Orthogonal transformation also find theRank, index,signature& nature.	CO2	L5	**10 M**
UNIT – III					
5.	**a)**	Verify Rolle's theorem for$f(x) = log\left(\frac{x^2+ab}{x(a+b)}\right)$ in (a, b).	CO3	L3	**5 M**
	b)	Verify Lagrange's mean value theorem for the functions$f(x) = (x-1)(x-2)(x-3)$ in $(0, 4)$	CO3	L3	**5 M**
(OR)					
6.	**a)**	Verify Cauchy's mean value theorem for the functions$sinx$&$cosx$ in $(0, \pi)$	CO3	L3	**5 M**
	b)	ObtainTaylors series expansion of $sinx$ in powers of $x - \frac{\pi}{4}$.	CO3	L3	**5 M**
UNIT – IV					
7.	**a)**	Find the first and second partial derivatives of $z = x^3 + y^3 - 3axy$	CO4	L1	**5 M**
	b)	Expand $x^2y + 3y - 2$ in powers of (x-1) and (y+2)	CO4	L3	**5 M**
(OR)					
8.	**a)**	If u= $\sqrt{x^2+y^2}$ and v =$\tan^{-1}\left(\frac{y}{x}\right)$, find $J\left(\frac{u,v}{x,y}\right)$	CO4	L1	**5 M**
	b)	Discuss the maximum and minimum values of $f(x, y) = 1 - x^2 - y^2$	CO4	L5	**5 M**
UNIT – V					
9.	**a)**	Find the value of the double integral$\int_1^2 \int_1^3 xy^2\,dxdy$	CO5	L1	**5 M**
	b)	Evaluate $\iint xydxdy$ over the positive quadrant of the circle $x^2 + y^2 = a^2$	CO5	L5	**5 M**
(OR)					
10.	**a)**	Evaluate the following double integrals by changing to polar co-ordinates $\int_0^\infty \int_0^\infty e^{-(x^2+y^2)}dxdy$	CO5	L5	**5 M**
	b)	Find the value of the Triple integral $\int_1^e \int_1^{\log y} \int_1^{e^x} \log z\, dz\, dx\, dy$	CO5	L1	**5 M**

***** End of the Question Paper *****

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B. Tech I Semester Regular Examinations

LINEAR ALGEBRA AND CALCULUS

(CIVIL, EEE, MECH, ECE, CSE, IT, AI&DS, AI&ML)

(Model Paper-II)

Time: 3 hours **Max Marks: 70M**

Note: 1. Question paper consists of two parts (**Part A and Part B**)
2. Answer all sub questions from **Part A**
3. Answer all questions from **Part B** with **either or choice**

		PART – A			**10 X 2 = 20 M**
			CO	**BTL**	**Marks**
I.	**a)**	Define rank of a matrix	CO1	L1	**2 M**
	b)	Solve $x+y+z=6\,, x-y+z=2\,,\ \ 2x-y+3z=9$	CO1	L3	**2 M**
	c)	Find the Eigen values of A^{-1} if A= $\begin{bmatrix} 2 & 3 & 4 \\ 0 & 4 & 2 \\ 0 & 0 & 3 \end{bmatrix}$	CO2	L1	**2 M**
	d)	Write the quadratic form corresponding to the symmetric matrix $\begin{bmatrix} 1 & 0 & -1 \\ 0 & -1 & 2 \\ -1 & 2 & 1 \end{bmatrix}$	CO2	L1	**2 M**
	e)	Explain why mean value theorem does not hold for $f(x)=x^{2/3}$ in [-1, 1].	CO3	L5	**2 M**
	f)	Write the Maclaurin's series of sin x	CO3	L1	**2 M**
	g)	If $x=u(1+v), y=v(1+u) then Show\ that \frac{\partial(x,y)}{\partial(u,v)}=1+u+v$	CO4	L2	**2 M**
	h)	Are $u=x\sqrt{1-x^2}\ \ ,v=2x$ functionally dependent?	CO4	L4	**2 M**
	i)	Evaluate $\int_0^2\int_0^x e^{x+y}\,dxdy$	CO5	L4	**2 M**
	j)	Evaluate $\int_0^1\int_0^2\int_1^2 x^2yz\,dxdydz$	CO5	L4	**2 M**
		PART – B			**5 X 10 = 50 M**
		UNIT - I			
1.	**a)**	Find the Inverse of the following matrices by Gauss-Jordan Method $\begin{bmatrix} 2 & 3 & 4 \\ 4 & 3 & 1 \\ 1 & 2 & 4 \end{bmatrix}$	CO1	L1	**5 M**
	b)	Determine the rank of the following matrices by reducing to the normal form $\begin{bmatrix} 8 & 1 & 3 & 6 \\ 0 & 3 & 2 & 2 \\ -8 & -1 & -3 & 4 \end{bmatrix}$	CO1	L5	**5 M**
		(OR)			
2.	**a)**	SolvetheFollowingEquationsbyGauss-Jacobi IterativeMethod10x+y-z=11.19 , x+10y+z=28.08, -x+y+10z=35.61	CO1	L3	**5 M**
	b)	Investigate the values of λ and μ so that the equations 2x + 3y + 5z = 9,7x + 3y – 2z = 8, 2x + 3y + λz = μ have (i) no solution (ii) a unique solution (iii) an infinite number of solutions.	CO1	L4	**5 M**

UNIT – II					
3.	**a)**	Verify the Cayley – Hamilton theorem for the following matrices also find A^{-1} for $\begin{bmatrix} 4 & 6 & 6 \\ 1 & 3 & 2 \\ -1 & -4 & -3 \end{bmatrix}$	CO2	L2	**5 M**
	b)	Determine the Eigen values and Eigen vectors of the following matrices $\begin{bmatrix} 1 & 1 & 3 \\ 1 & 5 & 1 \\ 3 & 1 & 1 \end{bmatrix}$	CO2	L5	**5 M**
(OR)					
4.		Reduce the Quadratic form $3x^2 + 5y^2 + 3z^2 - 2yz + 2xz - 2xy$ to canonical form by Orthogonal transformation also find theRank, index,signature& nature.	CO2	L5	**10 M**
UNIT – III					
5.	**a)**	Verify Rolle's theorem for$f(x) = sinx \,/\, e^x$ in $(0, \pi)$	CO3	L3	**5 M**
	b)	P.T if$(0 < a < b < 1)$, $\frac{b-a}{1+b^2} < tan^{-1}b - tan^{-1}a < \frac{b-a}{1+a^2}$. Hence S.T $\frac{\pi}{4} + \frac{3}{25} < tan^{-1}\frac{4}{3} < \frac{\pi}{4} + \frac{1}{6}$	CO3	L2	**5 M**
(OR)					
6.	**a)**	$If f(x) = \sqrt{x}$ and g(x) = $\frac{1}{\sqrt{x}}$ hence P. T 'c' of Cauchy's mean value theorem is Geometric mean between a & b	CO3	L2	**5 M**
	b)	Using Maclaurin's series expand the function e^x	CO3	L3	**5 M**
UNIT – IV					
7.	**a)**	If $z = f(x + ct) + \varphi(x - ct)$, prove that $\frac{\partial^2 z}{\partial t^2} = c^2 \frac{\partial^2 z}{\partial x^2}$	CO4	L1	**5 M**
	b)	Expand $e^x \log(1 + y)$ in powers of x and y	CO4	L3	**5 M**
(OR)					
8.	**a)**	If $u = x^2 - y^2$, $v = 2xy$ where $x = rcos\theta$, $y = rsin\theta thenshowthat \frac{\partial(u,v)}{\partial(r,\theta)} = 4r^3$	CO-4	L2	**5 M**
	b)	Find the maximum and minimum values of $x + y + z$ subject to $\frac{1}{x} + \frac{1}{y} + \frac{1}{z} = 1$	CO-4	L1	**5 M**
UNIT – V					
9.	**a)**	Evaluate the double integral$\int_0^5 \int_0^{x^2} x(x^2 + y^2)dxdy$	CO5	L5	**5 M**
	b)	Evaluate$\int\int r^3\, drd\theta$ over the area bounded by the circles $r = 2\sin\theta$ and $r = 4\sin\theta$	CO5	L5	**5 M**
(OR)					
10.	**a)**	Evaluate the following double integrals by changing the order of integration$\int_0^{4a} \int_{\frac{x^2}{4a}}^{2\sqrt{ax}} dy\, dx$	CO5	L5	**5 M**
	b)	Find$\int_0^{\log 2} \int_0^{x} \int_0^{x+\log y} e^{x+y+z} dxdydz$	CO5	L1	**5 M**

***** End of the Question Paper *****